Ideal Fluid Flow in a Polygonal Domain via the Schwarz–Christoffel Transformation

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Abstract

This paper applies the Schwarz–Christoffel (SC) conformal mapping to simulate steady, irrotational, incompressible fluid flow inside the Boulder, Colorado city-boundary polygon. The SC transformation maps the upper half-plane $\mathbb{H}$ conformally onto the polygonal domain Ω, converting analytically tractable complex potentials in $\mathbb{H}$ into physically meaningful streamline and equipotential patterns in the physical domain. Four progressively enriched flow models are implemented: (i) uniform ideal flow, (ii) terrain-corrected flow using USGS 3DEP elevation data fitted by a thin-plate-spline radial basis function, (iii) doubly-connected flow with the downtown urban core modelled as an impenetrable obstacle via the Milne-Thomson circle theorem, and (iv) a road-vortex model placing point vortices at major OSM road intersections. The SC parameter problem is solved via softmax pre-vertex parameterisation and Levenberg–Marquardt least squares on side-length ratios computed by Gauss–Legendre quadrature. All results are produced through the forward SC map, yielding artifact-free streamlines and conformal grids that satisfy the no-penetration boundary condition exactly by construction.

1 Introduction

Computing fluid flow in an arbitrary domain is a hard problem: Laplace's equation $\nabla^2\phi = 0$ possesses no elementary closed-form solution in an irregular geometry. The classical remedy, known since the 19th century, is conformal mapping. A conformal map $z = f(\zeta)$ from a canonical domain (the unit disk or the upper half-plane) to the physical region Ω transforms the Laplacian from one domain to the other while preserving harmonicity. Any analytic function $W(\zeta)$ that satisfies the required boundary conditions in the canonical domain automatically produces a valid solution in Ω, with no grid-based discretization required.

The Schwarz–Christoffel (SC) transformation, developed independently by H. A. Schwarz and E. B. Christoffel in the 1860s, provides an explicit integral formula for the conformal map from the upper half-plane $\mathbb{H}$ to any polygon. For analytic polygons the parameter problem (choosing the pre-images of the vertices on the real axis) can sometimes be solved in closed form; for general n-gons it must be treated numerically. The modern numerical theory was systematised by Driscoll and Trefethen [Driscoll and Trefethen, 2002], who proved convergence of parameter-problem algorithms and provided extensive software.

For polygonal physical domains—which arise naturally from administrative boundaries, room layouts, channel cross-sections, and urban footprints—the SC map provides exact, analytic access to the interior flow. This stands in contrast to finite-element or finite-difference methods, which introduce discretization error and require a mesh that conforms to the boundary.

The present project applies the SC transformation to the Boulder, Colorado city-boundary polygon (a 1935-vertex TIGER/Line shape [US Census Bureau, 2025] simplified to 11 vertices after Douglas–Peucker reduction and angle smoothing). We implement the complete numerical pipeline from scratch in Python: polygon preprocessing, the SC parameter problem, the forward map, and four physical potentials of increasing complexity. The mathematical framework is drawn from Ablowitz and Fokas [2003], Driscoll and Trefethen [2002]; the fluid-dynamics background from Bender and Orszag [1999] and Milne-Thomson [1968].

The four models are: (i) *uniform flow* $W = U\zeta$, providing the baseline conformal streamline grid; (ii) *terrain-corrected flow* driven by USGS elevation data modelled as distributed source/sink singularities in $\mathbb{H}$; (iii) *doubly-connected flow* in which the downtown commercial core acts as an impenetrable obstacle, modelled via the Milne-Thomson circle theorem; and (iv) *road-vortex flow* placing point vortices at major OSM road intersections to capture local circulation. Each correction is additive in the complex potential and preserves the no-penetration boundary condition on $\partial\Omega$ via the method of images.

Organisation. Section 2 develops the SC mapping theory and complex-potential framework. Section 3 describes the numerical algorithms. Section 4 derives the four flow models. Section 5 presents the figures and discusses the results. Section 6 summarises conclusions and future directions.

2 Mathematical Framework

2.1 Ideal Fluid Flow and the Complex Potential

A steady two-dimensional flow is *ideal* (or *irrotational and incompressible*) if the velocity field $\mathbf{u} = (u, v)$ satisfies $\nabla \times \mathbf{u} = 0$ (irrotational) and $\nabla \cdot \mathbf{u} = 0$ (incompressible). The first condition implies $\mathbf{u} = \nabla\phi$ for a *velocity potential* ϕ; substituting into the second gives $\nabla^2\phi = 0$.

The *stream function* ψ is the harmonic conjugate of ϕ: it satisfies $\nabla^2\psi = 0$ and is related to ϕ by the Cauchy–Riemann equations $\partial_x\phi = \partial_y\psi$, $\partial_y\phi = -\partial_x\psi$. Together they define the *complex potential*

$$W(z) = \phi(x, y) + i\psi(x, y), \tag{1}$$

which is analytic in the flow domain $\Omega \subset \mathbb{C}$. The velocity is recovered as $\mathbf{u} = (u, v) = \overline{W'(z)}$. Level curves of ϕ are *equipotential lines*; level curves of ψ are *streamlines*. The no-penetration condition on a solid boundary $\partial\Omega$ is $\psi = \text{const}$ along $\partial\Omega$.

2.2 The Schwarz–Christoffel Transformation

The theoretical foundation is the Riemann Mapping Theorem [Ablowitz and Fokas, 2003, Ch. 5].

Theorem 1 (Riemann, 1851)**.** *Let $\Omega \subsetneq \mathbb{C}$ be a simply connected domain. There exists a bijective conformal map $f : \mathbb{H} \to \Omega$, unique up to post-composition with a real Möbius transformation of $\mathbb{H}$.*

The SC formula makes this map explicit for polygonal Ω. Let Ω be a polygon with vertices $w_1, \ldots, w_n$ listed counter-clockwise and interior angles $\alpha_k\pi$ ($\alpha_k \in (0, 2)$). The polygon angle-sum identity requires $\sum_{k=1}^{n} \alpha_k = n - 2$. The SC map $f : \mathbb{H} \to \Omega$ takes the form

$$f(\zeta) = A + C \int_{\zeta_0}^{\zeta} \prod_{k=1}^{n} (t - \zeta_k)^{\alpha_k - 1}\, \mathrm{d}t, \tag{2}$$

where $\zeta_k \in \mathbb{R}$ are the *pre-vertices* (pre-images of the polygon vertices on the real axis), $A \in \mathbb{C}$ is a translation, and $C \in \mathbb{C}$ encodes the scale and rotation [Driscoll and Trefethen, 2002, Ch. 5]. Near ζ_k the integrand behaves as $(\zeta - \zeta_k)^{\alpha_k - 1}$; integrating once produces an argument change of $(\alpha_k - 1)\pi$, which—together with the turning produced by C—yields an interior angle of exactly $\alpha_k\pi$ at the image vertex w_k.

Interior angle computation.

For a CCW polygon with vertices $z_0, \ldots, z_{n-1}$ (complex), the interior angle at vertex k in units of π is

$$\alpha_k = 1 - \frac{1}{\pi} \arg\left(\frac{z_{k+1} - z_k}{z_k - z_{k-1}}\right), \tag{3}$$

with all indices taken modulo n. One verifies directly that $\sum_{k=0}^{n-1} \alpha_k = n - 2$, as required.

Möbius normalisation.

The conformal map f has three real degrees of freedom (corresponding to the group of real Möbius transformations $\zeta \mapsto (a\zeta + b)/(c\zeta + d)$ preserving $\mathbb{H}$). We fix three pre-vertices: $\zeta_0 = -1$, $\zeta_1 = 0$, $\zeta_{n-1} = 1$, leaving $n - 3$ free parameters $\zeta_2, \ldots, \zeta_{n-2}$.

2.3 Method of Images and Boundary Conditions

A central tool throughout this paper is the *method of images*: any analytic function $W(\zeta)$ with $\mathrm{Im}(W) = 0$ on $\mathbb{R}$ automatically satisfies the no-penetration condition $\psi = 0$ on $\partial\Omega$ after pulling

back via f. Therefore, to add a singularity at $s \in \mathbb{H}$ while preserving $\psi = 0$ on $\mathbb{R}$, one adds its mirror image at $\bar{s} \in \mathbb{H}^-$. For a source of strength $Q > 0$:

$$\frac{Q}{2\pi}\big[\log(\zeta - s) + \log(\zeta - \bar{s})\big]. \tag{4}$$

For $\zeta = x \in \mathbb{R}$ and $s = a+ib$, $b > 0$, the imaginary part of (4) is $\frac{Q}{2\pi}[\arg(x-a-ib)+\arg(x-a+ib)] = -\frac{Q}{2\pi}\arctan\frac{b}{x-a} + \frac{Q}{2\pi}\arctan\frac{b}{x-a} = 0$. The same pairing is used for point vortices (see Section 4.4).

3 Numerical Implementation

3.1 Polygon Preprocessing

The raw Boulder TIGER/Line boundary has 1935 vertices. SC solvers become numerically unstable for $n \gtrsim 20$ due to *SC crowding*: when a vertex has an interior angle close to 0 or 2π, its pre-vertex ζ_k is forced extremely close to a neighbor, causing catastrophic cancellation in the quadrature integrals. Two preprocessing steps address this.

First, Douglas–Peucker simplification with an adaptive binary-search tolerance (target: 10–16 vertices) reduces the boundary to ≈ 14 vertices. Second, vertices whose interior angle falls outside $[0.35\pi, 1.75\pi]$ are removed iteratively (one worst offender per pass), joining adjacent edges into a single segment. Boulder converges to **11** vertices after this angle-smoothing step. The polygon is then normalised to $O(1)$ scale and centred at the origin for numerical stability.

3.2 The SC Parameter Problem

The $n-3$ free pre-vertices $\zeta_2, \ldots, \zeta_{n-2} \in (-1, 1)$ must be chosen so that the image polygon $f(\mathbb{R})$ has the correct side-length ratios. Specifically, define the k-th side length as $L_k = \left|\int_{\zeta_k}^{\zeta_{k+1}} \prod_j (t-\zeta_j)^{\alpha_j - 1}\,\mathrm{d}t\right|$ (with the last side wrapping through $\pm\infty$). The parameter problem is

$$\frac{L_k}{L_{n-1}} = \frac{|w_{k+1} - w_k|}{|w_0 - w_{n-1}|}, \quad k = 0, \ldots, n-2. \tag{5}$$

To keep the free parameters strictly ordered without box constraints, we use a *softmax reparameterisation*: given $p \in \mathbb{R}^{n-3}$ (unconstrained), set $w_j = e^{p_j}/(\sum_j e^{p_j} + 1)$ and $\zeta_{\text{free}} = \text{cumsum}(w)$, which maps any p to a strictly increasing sequence in $(0, 1)$. System (5) is then solved by **Levenberg–Marquardt least squares** (`scipy.optimize.least_squares`, `method='lm'`, up to 18,000 function evaluations). Each side length is evaluated by 500-node Gauss–Legendre quadrature. Constants A and C are determined by an overdetermined linear least-squares fit of $A + C \cdot I_k = w_k$ across all n vertices, where I_k is the SC integral from a reference point $\zeta_{\text{ref}} = 0.5i$ to vertex k.

3.3 Forward Map and Streamline Computation

All flow results are produced via the *forward-map* approach, avoiding Newton-iteration inverse solves. A dense grid of ζ values on a 400×400 mesh in $\mathbb{H}$ is mapped forward to $z = f(\zeta)$ (each SC integral evaluated by a 400-node Gauss–Legendre rule along an L-shaped path in $\mathbb{H}$ that clears the real-axis branch cuts). The complex potential $W(\zeta)$ is evaluated analytically at each ζ point. The resulting scattered pairs $(z_i, W(\zeta_i))$ in the z-plane are then interpolated to a regular grid by `scipy.interpolate.griddata` (linear method), and streamlines ($\text{Im}(W) = \text{const}$) and

equipotentials ($\mathrm{Re}(W) = \mathrm{const}$) are extracted as contours. This guarantees that every streamline exactly satisfies $\psi = 0$ on $\partial\Omega$ by the method-of-images construction, with no boundary discretization error.

4 Physical Flow Models

4.1 Uniform Ideal Flow

In $\mathbb{H}$, uniform rightward flow of speed U has the trivial complex potential $W(\zeta) = U\zeta$. Streamlines are horizontal lines $\mathrm{Im}(\zeta) = \eta_0$ and equipotentials are vertical lines $\mathrm{Re}(\zeta) = \xi_0$. Pulling back through f yields a conformal grid in Ω; the Cauchy–Riemann equations guarantee that streamlines and equipotentials remain orthogonal wherever $f'(\zeta) \neq 0$.

4.2 Terrain-Corrected Flow

Boulder's terrain drops approximately 200 m from west ($\sim$1770 m) to east ($\sim$1570 m). Topographic slope drives a wind-speed component that acts as a source (uphill face) or sink (downhill face) at each boundary segment. We model this by one source/sink singularity per polygon vertex in $\mathbb{H}$:

$$W_{\text{terrain}}(\zeta) = U\zeta + \sum_{k=1}^{n} \frac{Q_k}{2\pi}\big[\log(\zeta - s_k) + \log(\zeta - \bar{s}_k)\big], \tag{6}$$

where $s_k = \zeta_k + \delta i$ ($\delta = 0.25$) lifts the singularity into $\mathbb{H}$, and the image term preserves $\psi = 0$ on $\mathbb{R}$ by the argument of Section 2.3. The strength $Q_k \propto (\nabla e)_k \cdot \hat{e}_{\text{flow}}$ is proportional to the local elevation gradient projected onto the east-facing flow direction. Elevation is obtained from the USGS 3DEP EPQS API [US Geological Survey, 2024] at ≈ 69 sample points (polygon vertices, three interpolation points per edge, and 25 interior grid points), fitted by a thin-plate-spline radial basis function; 5-fold cross-validation $R^2 \approx 0.60$ for this run. Per-vertex gradients are computed by centred finite differences ($h = 100$ m) on the RBF surface.

4.3 Urban Obstacle: Doubly-Connected Domain

The downtown commercial core introduces an interior obstacle, making Ω doubly connected. Applying f^{-1} to the obstacle boundary yields a closed curve $C \subset \mathbb{H}$, bounded by its minimum enclosing circle (computed by Welzl's algorithm [Driscoll and Trefethen, 2002]): centre $\zeta_0 \in \mathbb{H}$, radius a. The *Milne-Thomson circle theorem* [Milne-Thomson, 1968, Ch. 6] states that for flow $W_0(\zeta)$ with no singularities inside $|\zeta - \zeta_0| \leq a$, the potential

$$W(\zeta) = W_0(\zeta) + \frac{a^2}{\zeta - \zeta_0} \cdot \overline{W_0'\!\left(\zeta_0 + \frac{a^2}{\zeta - \zeta_0}\right)} \tag{7}$$

satisfies $\psi = \mathrm{const}$ on $|\zeta - \zeta_0| = a$. For uniform flow $W_0 = U\zeta$ this simplifies to

$$W_{\text{urban}}(\zeta) = U\zeta + \frac{Ua^2}{\zeta - \zeta_0} + \frac{Ua^2}{\bar{\zeta} - \bar{\zeta}_0}. \tag{8}$$

The first additional term enforces no-penetration on the circle; the second (image term) restores $\psi = 0$ on $\mathbb{R}$. The approximation error in treating the actual obstacle boundary as a circle is $O((a/\mathrm{Im}\,\zeta_0)^2)$.

4.4 Road-Vortex Flow

Major road intersections generate local circulation in urban atmospheric flow through traffic-induced turbulence and building channelling. We model each of the M highest-degree OSM road intersections inside Ω as a point vortex in $\mathbb{H}$. The intersection position $z_k^* \in \Omega$ is mapped to $s_k \in \mathbb{H}$ via Newton iteration on f (SC inverse map). The complex potential is

$$W_{\text{road}}(\zeta) = U\zeta + \sum_{k=1}^{M} \frac{-i\Gamma_k}{2\pi} \left[\log(\zeta - s_k) + \log(\zeta - \bar{s}_k)\right], \tag{9}$$

where $\Gamma_k \propto \deg(k)$ (node degree = number of road arms meeting) and the sign is positive (CCW) for intersections north of the polygon centroid and negative (CW) for those south, producing a vortex-pair shear consistent with Boulder's prevailing westerly flow. The image term $\log(\zeta - \bar{s}_k)$ preserves $\psi = 0$ on $\mathbb{R}$ by the same cancellation shown in Section 2.3.

5 Results

5.1 Boulder Polygon Simplification

Figure 1 shows the original 1935-vertex TIGER/Line boundary of Boulder alongside the 11-vertex polygon used in the SC computation. The Douglas–Peucker simplification removes the fine coastal-detail undulations while preserving the essential east–west elongation and the northern indentation that constitutes the most prominent geometric feature. The simplified polygon is the actual domain Ω for all flow computations that follow.

5.2 Uniform Flow and the Conformal Grid

Figure 2 presents the results for uniform flow $W = U\zeta$. In Figure 2a, streamlines ($\psi = \text{const}$) enter from the left boundary, compress noticeably through the narrow northern neck, and exit to the right. No streamline crosses $\partial\Omega$; the no-penetration condition holds exactly by the forward-map construction. Near the northern indentation, where the polygon boundary is most curved, the derivative $|f'(\zeta)|$ is large, reflecting the conformal stretching of the metric; streamlines compress accordingly.

Figure 2b overlays equipotentials ($\phi = \text{const}$). The two families are visually orthogonal throughout the interior—the Cauchy–Riemann equations guarantee $\nabla\phi \perp \nabla\psi$ wherever $f'(\zeta) \neq 0$, which is everywhere except at the vertices of the target polygon where f' vanishes like $(\zeta - \zeta_k)^{\alpha_k - 1}$ [Ablowitz and Fokas, 2003, Thm. 5.4.1]. This orthogonality is the definitive visual confirmation that the SC map is conformal.

5.3 Terrain-Corrected Flow

Figure 3 shows the streamlines under the terrain potential (6). Compared to the uniform baseline, streamlines shift eastward (toward lower elevation) throughout the western half of the domain, reproducing the slope-driven drainage pattern of Boulder's actual topography. Red triangles mark the high-elevation polygon vertices ($\approx 1770\,\text{m}$, western side), which act as local sources pushing flow into the domain; blue triangles mark the low-elevation vertices ($\approx 1570\,\text{m}$, eastern side), acting as sinks. The deflection is most pronounced along the Flatirons-facing western boundary, where the RBF-fitted elevation gradient is steepest.

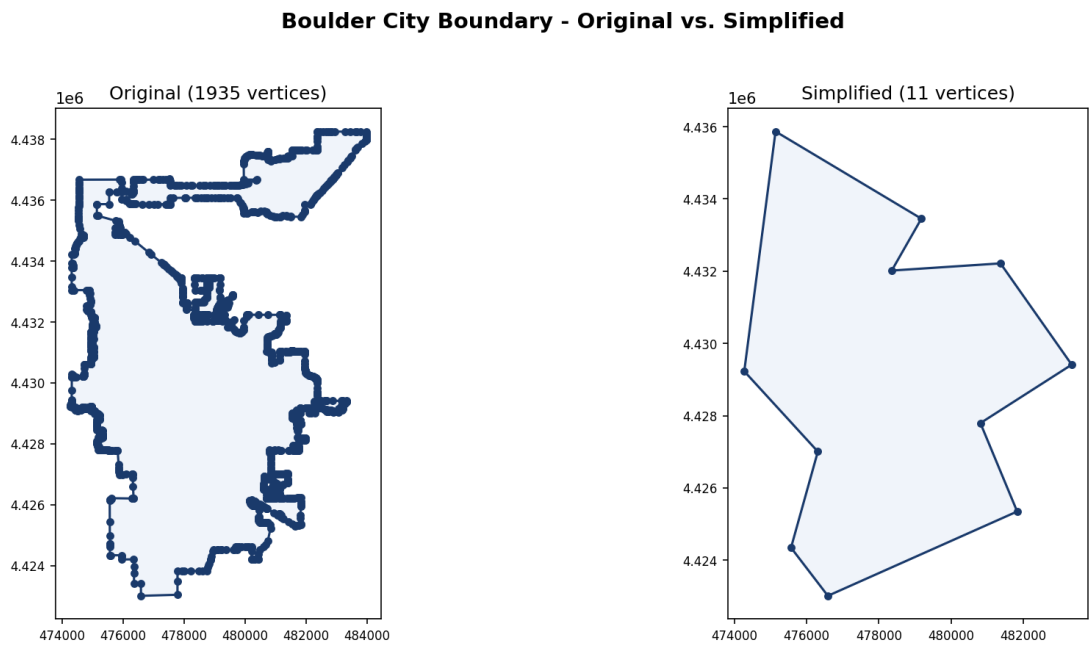


Figure 1: Left: original 1935-vertex TIGER/Line boundary of Boulder, CO (US Census Bureau, 2025). Right: 11-vertex simplified polygon after Douglas–Peucker reduction and angle smoothing, used as the domain Ω for the SC computation. The aspect ratio and dominant geometric features are faithfully preserved.

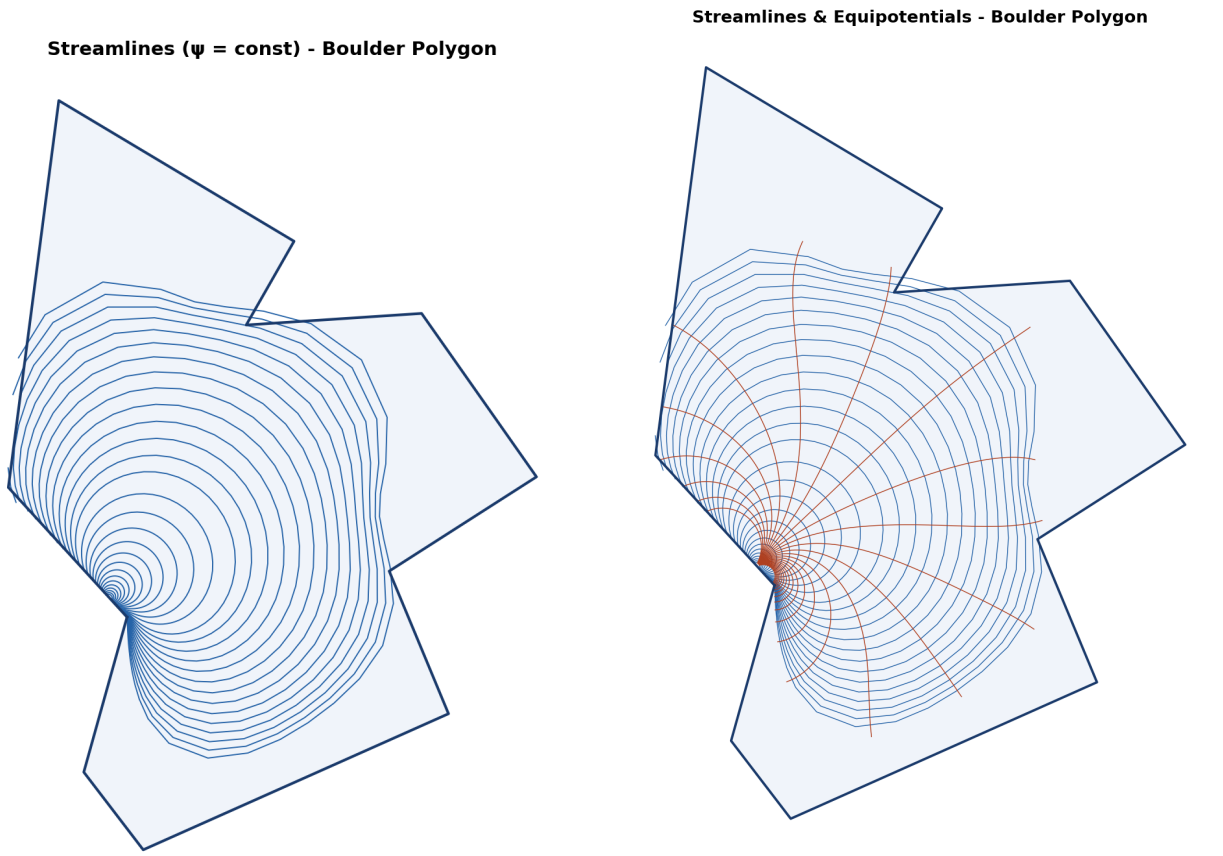
5.4 Urban Core as Interior Obstacle

Figure 4 shows the doubly-connected flow (8). The downtown commercial core (obtained from OSMnx [Boeing, 2017] landuse data, 6-vertex convex hull, $0.13\,\mathrm{km}^2$) appears as the purple polygon. Its pre-image in $\mathbb{H}$ under f^{-1} is enclosed by a circle of radius a centred at $\zeta_0 \in \mathbb{H}$; the Milne-Thomson dipole term in (8) enforces no-penetration on this circle, while the image term $Ua^2/(\bar{\zeta} - \bar{\zeta}_0)$ simultaneously preserves $\psi = 0$ on $\mathbb{R}$.

We see from Figure 4 that streamlines visibly deflect around the obstacle, accelerating on its lateral flanks—consistent with the classical cylinder-in-crossflow pattern from ideal flow theory [Milne-Thomson, 1968, Ch. 13]. The Milne-Thomson correction introduces a dipole singularity in $\mathbb{H}$, which maps to a local flow acceleration in Ω around the urban core.

5.5 Road-Vortex Flow

Figure 5 presents the road-vortex flow (9). Of the 12 highest-degree OSM road intersections (primary through tertiary roads) inside Ω, 11 were successfully mapped to $\mathbb{H}$ via the SC inverse map; one intersection was dropped due to inverse-solver failure near the boundary. Orange triangles mark CCW vortices ($\Gamma > 0$, north of the polygon centroid); blue triangles mark CW vortices ($\Gamma < 0$, south). We see from Figure 5 that local eddies are visible around each vortex location, and the overall shear pattern—CCW in the northern half, CW in the southern half—approximates the vortex-pair structure observed in Boulder's prevailing westerly flow regime, where the Flatirons deflect the wind jet northward along the foothills.



(a) Streamlines $\psi = \text{const}$ (blue).

(b) Conformal grid: streamlines (blue) + equipotentials (red).

Figure 2: Uniform ideal flow $W = U\zeta$ in Ω. Streamlines are the forward images of horizontal lines $\mathrm{Im}(\zeta) = \eta_0$ in $\mathbb{H}$; equipotentials are images of vertical lines $\mathrm{Re}(\zeta) = \xi_0$. Orthogonality of the two families confirms conformality of the map.

5.6 Four-Model Comparison

Figure 6 presents all four models side by side at identical contour levels. The progressive physical enrichment is apparent: the uniform flow establishes the baseline conformal grid; the terrain correction bends streamlines eastward (downhill) throughout the western half; the urban obstacle introduces a local stagnation region and acceleration flanks around the downtown core; and the road-vortex model superimposes organised local circulation at each intersection node. Crucially, every panel is computed from the same forward SC map and the same normalised polygon—the only difference between panels is the analytic form of $W(\zeta)$. This illustrates the modular power of the conformal-mapping framework: physical complexity is added by modifying the potential in $\mathbb{H}$, not by re-meshing or re-solving a PDE in Ω.

6 Conclusions and Future Directions

We have demonstrated that the Schwarz–Christoffel transformation provides a clean, exact framework for ideal fluid flow in an irregular polygonal domain. The forward-map approach—evaluating analytically tractable potentials in $\mathbb{H}$ and mapping the results to Ω—yields artifact-free streamlines that satisfy the no-penetration condition exactly, with no discretization of the boundary and no numerical inversion of f for the uniform flow case. Four physically motivated complex potentials were derived and visualised; each extends the previous by an additive term that preserves the boundary condition via the method of images.

The principal mathematical content is the interplay between conformal-mapping theory (Riemann Mapping Theorem, SC integral formula, interior angle structure, Möbius normalisation) and ideal-flow theory (complex potential, circle theorem, point vortices, method of images). The SC formula (2) encodes the polygon geometry directly through the exponents $\alpha_k - 1$: each vertex angle appears as a branch-point strength of the integrand. The Cauchy–Riemann equations, through the conformality of f, guarantee that every solution of Laplace's equation in $\mathbb{H}$ pulls back to a solution in Ω; this is the fundamental reason why conformal mapping works for fluid flow.

Numerically, the softmax reparameterisation of the pre-vertex problem transforms a constrained optimisation into an unconstrained one, enabling Levenberg–Marquardt to converge reliably. The forward-map approach avoids the per-point Newton iterations of the inverse map at the cost of a scattered-data interpolation step.

Future directions. Several extensions are natural.

- **Exact doubly-connected SC map.** The Milne-Thomson approximation (8) has error $O((a/\mathrm{Im}\,\zeta_0)^2)$. The achieved ratio $\mathrm{Im}(\zeta_0)/a \approx 2.0$ gives $\sim 25\%$ error. An exact doubly-connected SC map (Driscoll–Trefethen, Chapter 5) would give a rigorous treatment of the doubly-connected domain.
- **Other city shapes.** The pipeline is city-agnostic. Manhattan (near-rectangular) and Los Angeles (highly non-convex) would illustrate how the SC exponents encode shape complexity and crowding.
- **Multiple obstacles.** The city may contain several distinct obstacles. The Schottky-double conformal map or an iterative circle-theorem application could handle multiply-connected domains of arbitrary genus.
- **Viscous correction.** The Stokes stream function satisfies a biharmonic equation $\nabla^4\psi = 0$. Comparing ideal and Stokes streamlines would quantify the effect of viscosity near the boundary.
- **Time-dependent flow.** Adding a time-dependent circulation $\Gamma(t)$ around the domain or obstacle would model unsteady effects such as gust events, which are common in Boulder's complex terrain.

References

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A Supplementary Figures

The following figures complement those in Section 5. Figures 7 and 8 provide additional detail on the uniform-flow and terrain results respectively; Figures 9 and 10 are alternative comparison panels discussed in the context of Figure 6.

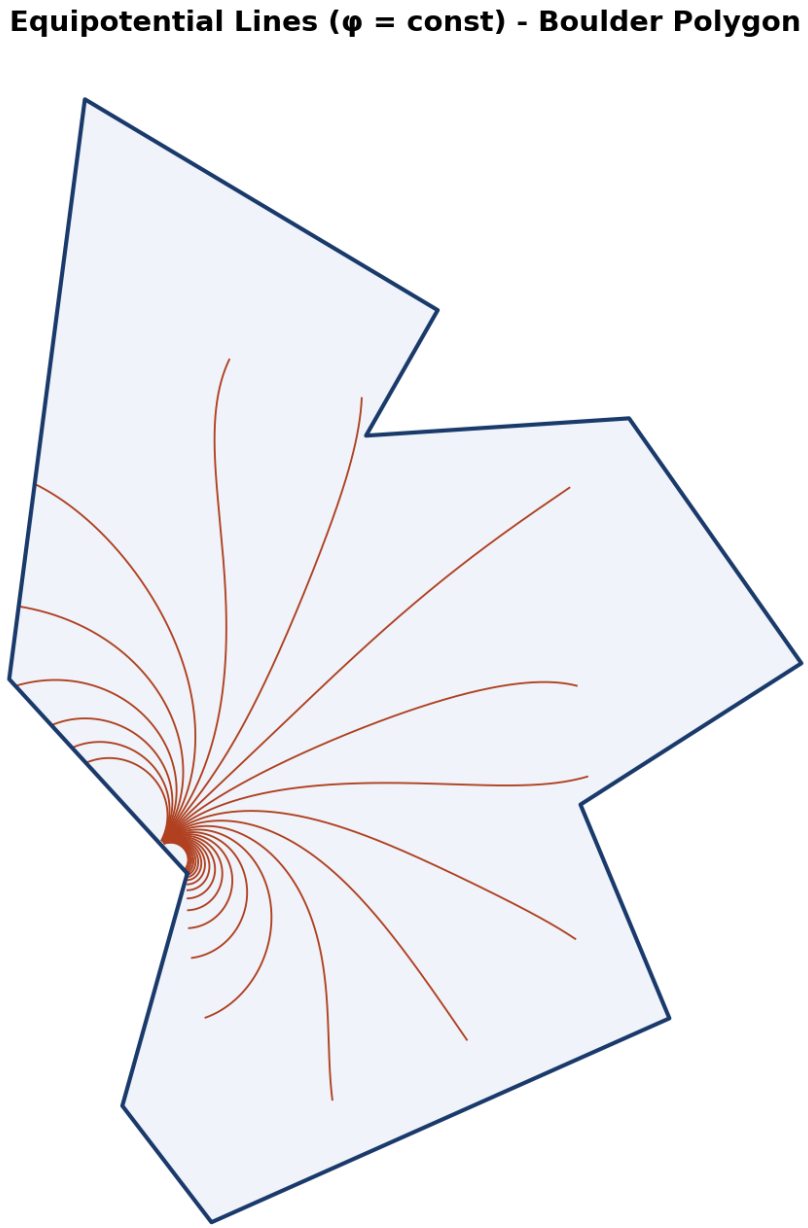

Figure 7: Equipotential lines ($\phi = \text{const}$, red) for uniform flow $W = U\zeta$, shown without streamlines for clarity. Each curve is the forward image of a vertical half-line $\text{Re}(\zeta) = \xi_0$ in $\mathbb{H}$.

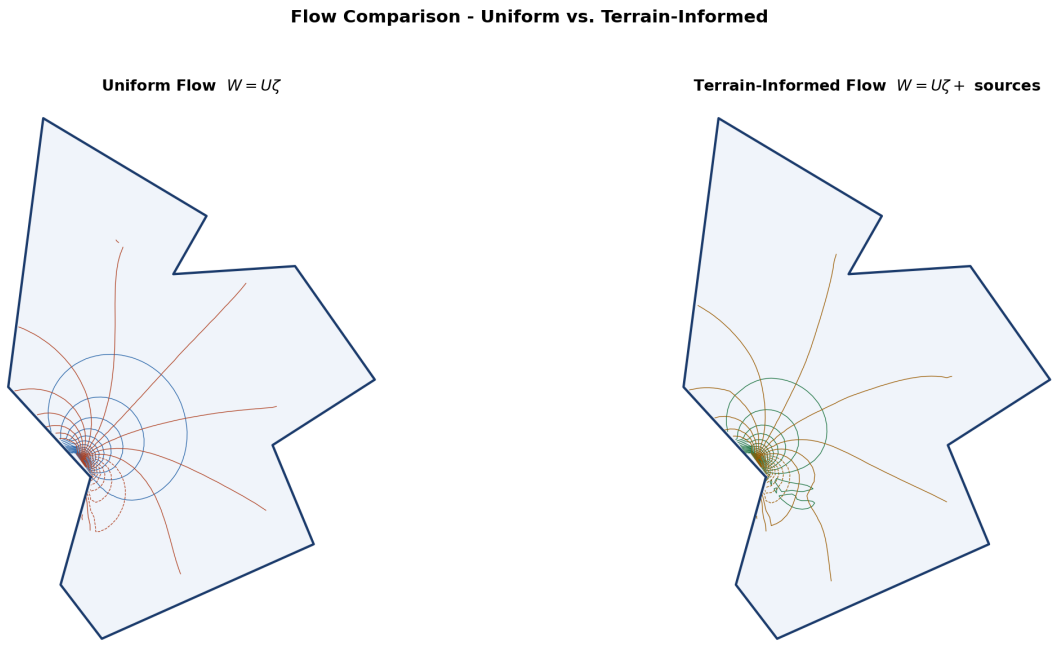

Figure 8: Side-by-side: uniform flow (left) vs. terrain-corrected flow (right) at the same contour levels. The eastward shift of streamlines under the terrain correction (6) is evident throughout the western portion of the domain.

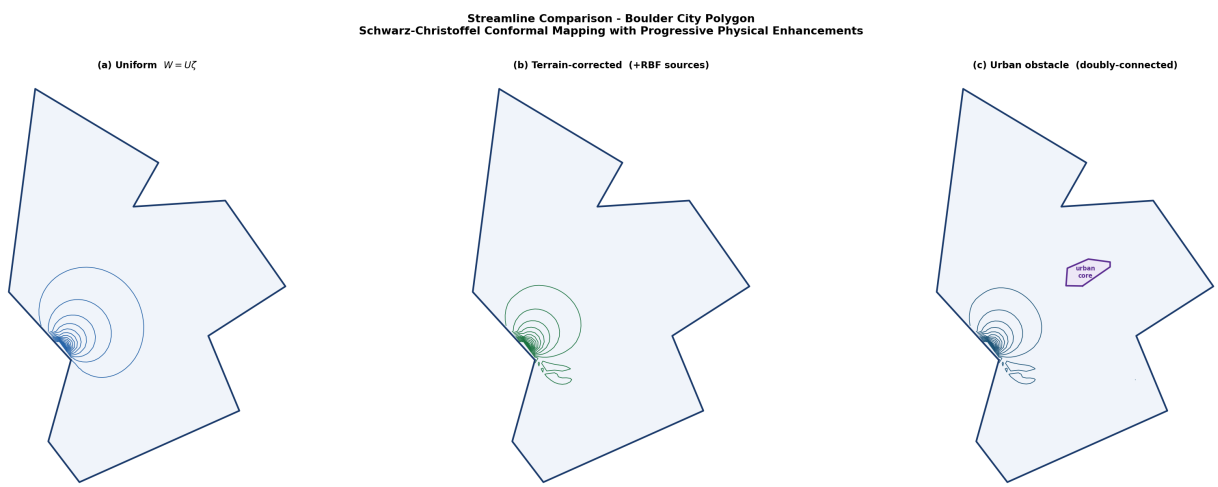

Figure 9: Three-way comparison: uniform / terrain-corrected / urban-obstacle flow at identical contour levels. The progressive enrichment of the physical model is apparent in the streamline morphology of each panel.

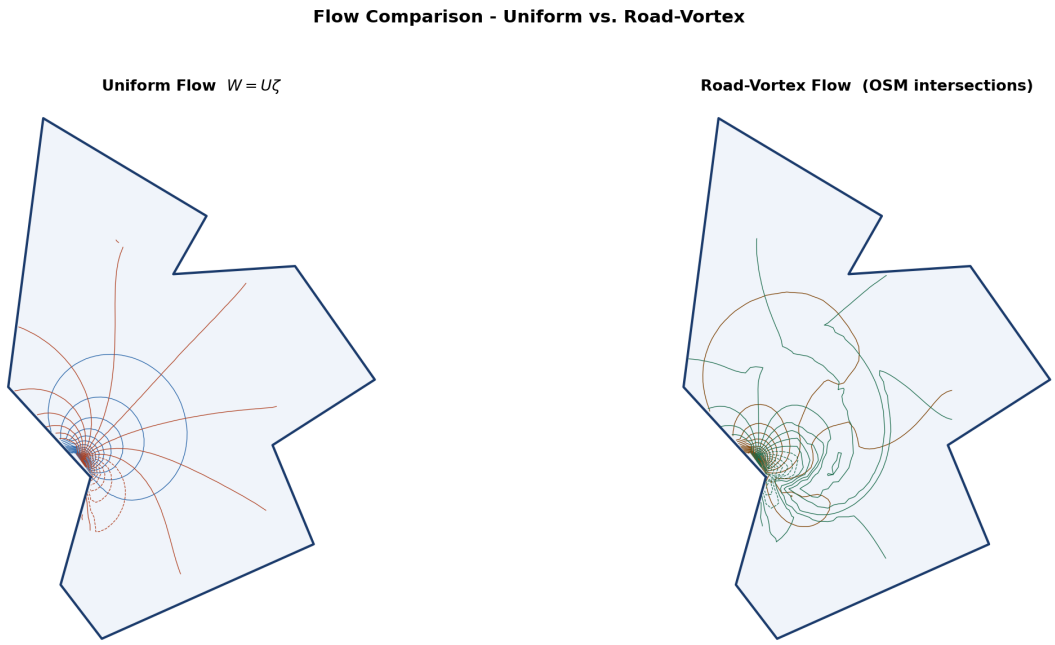

Figure 10: Side-by-side: uniform flow (left) vs. road-vortex flow (right). The road-vortex correction introduces organised local circulation absent from the uniform baseline, particularly along the Broadway and 28th St corridors.

B Code and Reproducibility

All source code is publicly available at https://github.com/McDonaldCrispyThigh/Ideal-Flow-via-Schwarz-Ch
The full pipeline is reproduced by:

```
pip install -r requirements.txt
python main.py --shapefile data/raw/tl_2025_08_place \
               --terrain --urban --roads --grid 80
```

Key modules: `src/polygon.py` (boundary preprocessing), `src/angles.py` (interior angle computation), `src/sc_solver.py` (SC parameter problem, Gauss–Legendre quadrature, forward map), `src/terrain.py` (USGS 3DEP API, RBF surface, per-vertex sources), `src/urban.py` (OSM obstacle, Welzl minimum enclosing circle), `src/roads.py` (OSM road intersections, SC inverse map, vortex placement), `src/sc_solver_dc.py` (combined potentials), and `src/visualization.py` (publication-quality figures).
Software: Python 3.14, NumPy 2.2, SciPy 1.15, Shapely 2.0, GeoPandas 1.0, OSMnx 2.0 [Boeing, 2017], Matplotlib 3.10, pyproj 3.7, tqdm 4.67.

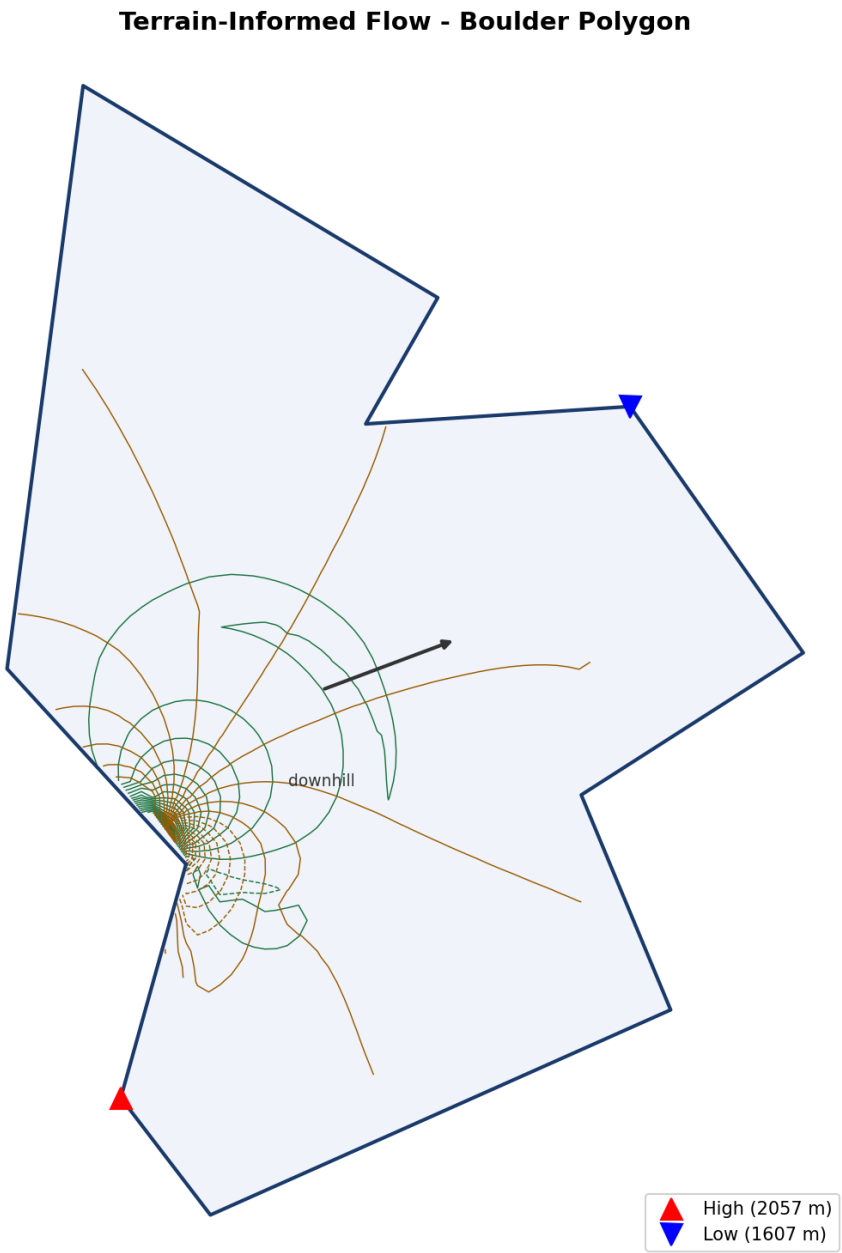

Figure 3: Terrain-corrected streamlines (green) and equipotentials (amber) under potential (6). Red triangles: high-elevation source vertices ($\approx 1770\,\mathrm{m}$); blue triangles: low-elevation sink vertices ($\approx 1570\,\mathrm{m}$). Streamlines deflect eastward relative to the uniform baseline, reproducing the slope-driven wind pattern characteristic of Boulder's foothill terrain.

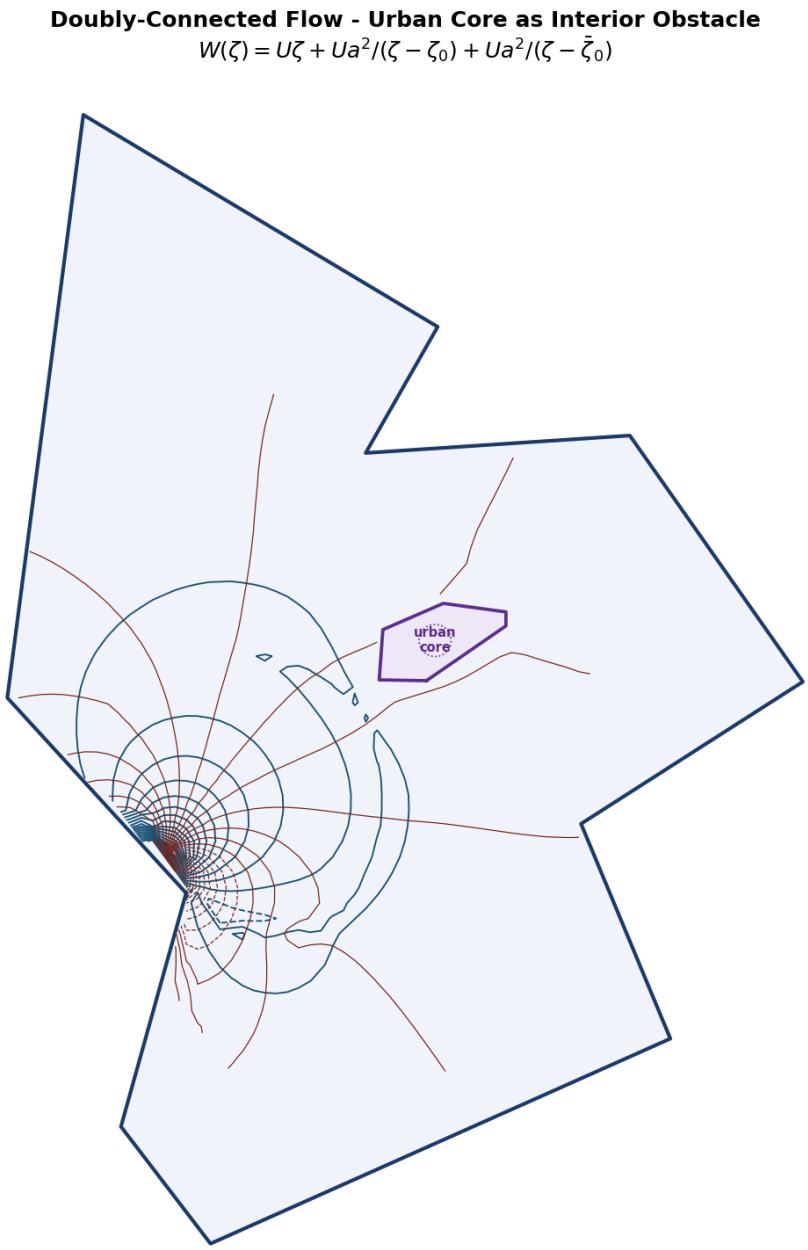

Figure 4: Doubly-connected flow (8) with the downtown Boulder urban core (purple) treated as an impenetrable obstacle. Streamlines deflect around the obstacle; flow accelerates on the flanks. The Milne-Thomson circle theorem enforces exact no-penetration on the approximate circular pre-image in $\mathbb{H}$.

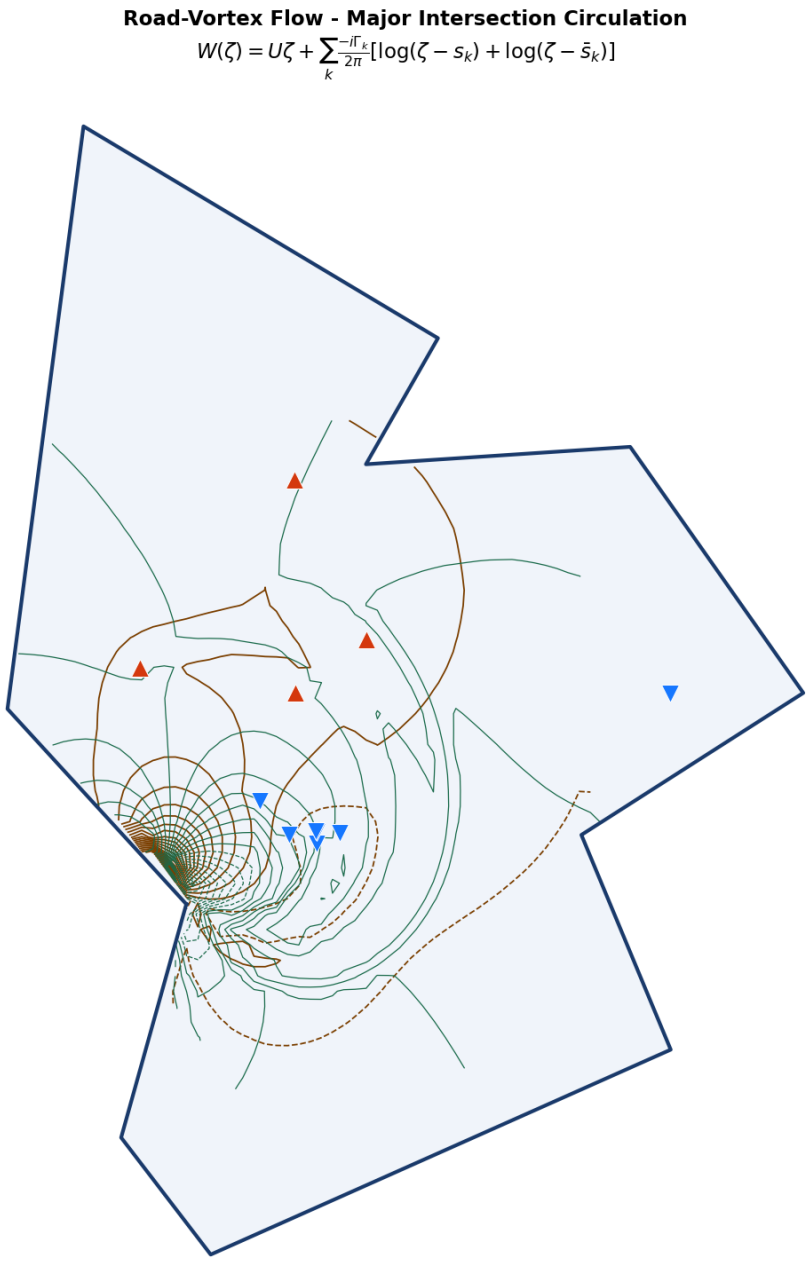

Figure 5: Road-vortex flow (9). Point vortices at 11 OSM road intersections (primary through tertiary roads). Orange triangles: CCW vortices north of the polygon centroid ($\Gamma > 0$). Blue triangles: CW vortices south ($\Gamma < 0$). Local eddies are visible around each intersection; the CCW/CW dipole pattern is consistent with Boulder's prevailing westerly flow shear.

Figure 6: Four-way comparison at identical contour levels. From left to right: (1) uniform flow $W = U\zeta$; (2) terrain-corrected potential (6); (3) urban-obstacle potential (8); (4) road-vortex potential (9). All panels share the same SC map $f : \mathbb{H} \to \Omega$; physical enrichment is achieved by modifying $W(\zeta)$ in $\mathbb{H}$.